

## Case Study 3

### FreshMart Supermarket - Sales & Inventory Analytics

**Industry: Retail**

**Modules: Reporting, Exploration, Dashboard, Story Making**

#### *Real-Time Business Scenario*

FreshMart is a supermarket chain with 35 outlets across 8 states in India. The chain sells groceries, fresh produce, dairy, bakery, and household items. The operations head has reported inconsistencies in stock levels — some outlets are overstocked while others face frequent stockouts. The sales team also wants to identify slow-moving products and top-selling categories to optimize purchasing decisions using IBM Cognos Analytics.

# Case Study Questions

## Question 1 [Reporting]

Create a Sales Summary Report grouped by Category and Sub-Category showing Total Sales, Total Profit, and Profit Margin %. add a calculated field for Profit Margin % = (Profit / Sales) \* 100. Format the report with a Header (FreshMart logo text + report date), Footer (page number + 'Confidential'), bold dark green headers, and a prompt for Outlet State selection.

### Sales Summary Report:

#### Screenshots

| FreshMart Sales Summary Report |                |                      | Date                  |                   |
|--------------------------------|----------------|----------------------|-----------------------|-------------------|
| Category                       | Sub-Category   | Total Sales          | Total Profit          | Profit Margin %   |
| <Category>                     | <Sub-Category> | <Total Sales >       | <Total Profit>        | <Profit Margin %> |
|                                | <Sub-Category> | <Total Sales >       | <Total Profit>        | <Profit Margin %> |
| <Category> - Total             |                | <Total(Total Sales)> | <Total(Total Profit)> |                   |
| <Category>                     | <Sub-Category> | <Total Sales >       | <Total Profit>        | <Profit Margin %> |
|                                | <Sub-Category> | <Total Sales >       | <Total Profit>        | <Profit Margin %> |
| <Category> - Total             |                | <Total(Total Sales)> | <Total(Total Profit)> |                   |
| Overall - Total                |                | <Total(Total Sales)> | <Total(Total Profit)> |                   |

Page 1 | Confidential

## Outlet State Selection Dropdown for Dynamic Report Filtering

Prompt

Provide values for the report you are about to run.

Outlet State

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Outlet State

Outlet State

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Delhi

Gujarat

Karnataka

Maharashtra

Rajasthan

Tamil Nadu

Telangana

OK

Cancel

## Sales Summary Report Showing Category, Sub-Category, Sales, Profit and Profit Margin (%)

| FreshMart Sales Summary Report |              |                   |           |                 | Apr 6, 2026 |
|--------------------------------|--------------|-------------------|-----------|-----------------|-------------|
| Category                       | Sub-Category | Total Sales Value | Profit    | Profit Margin % |             |
| Bakery                         | Sub-Bak      | 184224.7          | 62987.56  | 96.84%          |             |
| Bakery - Total                 |              | 184224.7          | 62987.56  |                 |             |
| Beverages                      | Sub-Bev      | 140234.25         | 16831.76  | 12.00%          |             |
| Beverages - Total              |              | 140234.25         | 16831.76  |                 |             |
| Dairy                          | Sub-Dai      | 342803.91         | 78233.85  | 65.14%          |             |
| Dairy - Total                  |              | 342803.91         | 78233.85  |                 |             |
| Fresh Produce                  | Sub-Fre      | 204621.02         | 27679.11  | 18.39%          |             |
| Fresh Produce - Total          |              | 204621.02         | 27679.11  |                 |             |
| Groceries                      | Sub-Gro      | 304360.97         | 53284.86  | 58.39%          |             |
| Groceries - Total              |              | 304360.97         | 53284.86  |                 |             |
| Household                      | Sub-Hou      | 274195.29         | 55265.08  | 81.74%          |             |
| Household - Total              |              | 274195.29         | 55265.08  |                 |             |
| Personal Care                  | Sub-Per      | 222574.55         | -2730.16  | 31.60%          |             |
| Personal Care - Total          |              | 222574.55         | -2730.16  |                 |             |
| Snacks                         | Sub-Sna      | 207591.33         | 39191.55  | 42.28%          |             |
| Snacks - Total                 |              | 207591.33         | 39191.55  |                 |             |
| Overall - Total                |              | 1880606.02        | 330743.61 |                 |             |

Page 1 | Confidential

## Question 2 [Reporting]

Build a Slow-Moving Products Report listing products where Quantity Sold is below 50 units in the last month. Apply Conditional Styling: Red for 0–20 units, Amber for 21–50 units. Add prompts for Category and Outlet City. Use font Trebuchet MS, size 11 for the report body. Include a count of total slow-moving products as a summary.

### Screenshots

#### Slow-Moving Products Report Layout with Product Details

| Slow-Moving Products Report  |                            |            |                |               | Date            |             |
|------------------------------|----------------------------|------------|----------------|---------------|-----------------|-------------|
| Product ID                   | Product Name               | Category   | Sub-Category   | Outlet City   | Quantity Sold   | Sale Date   |
| <Product ID>                 | <Product Name>             | <Category> | <Sub-Category> | <Outlet City> | <Quantity Sold> | <Sale Date> |
| <Product ID>                 | <Product Name>             | <Category> | <Sub-Category> | <Outlet City> | <Quantity Sold> | <Sale Date> |
| <Product ID>                 | <Product Name>             | <Category> | <Sub-Category> | <Outlet City> | <Quantity Sold> | <Sale Date> |
| <Total Slow-Moving Products> | Total Slow-Moving Products |            |                |               |                 |             |

1 | Confidential

#### Prompt Page for Selecting Category and Outlet City

Prompt

Provide values for the report you are about to run.

p\_Category

Fresh Produce

p\_City

Mumbai

OK

Cancel

Conditional Styling Applied – Low Sales Products Highlighted in Red (0–20 Units)

| Product ID | Product Name               | Category      | Sub-Category | Outlet City | Quantity Sold | Sale Date  |
|------------|----------------------------|---------------|--------------|-------------|---------------|------------|
| PRD0166    | Product_166                | Fresh Produce | Sub-Fre      | Mumbai      | 3             | 2024-12-07 |
| 1          | Total Slow-Moving Products |               |              |             |               |            |

Category and Outlet City Selection for Filtering Slow-Moving Products

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Bengaluru

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\*

Dairy

▼

Cancel

< Back

Next >

Finish

Conditional Styling Applied – Medium Low Sales Highlighted in Amber (21–50 Units)

| Product ID | Product Name               | Category | Sub-Category | Outlet City | Quantity Sold | Sale Date  |
|------------|----------------------------|----------|--------------|-------------|---------------|------------|
| PRD0187    | Product_187                | Dairy    | Sub-Dai      | Bengaluru   | 34            | 2024-12-13 |
| 1          | Total Slow-Moving Products |          |              |             |               |            |

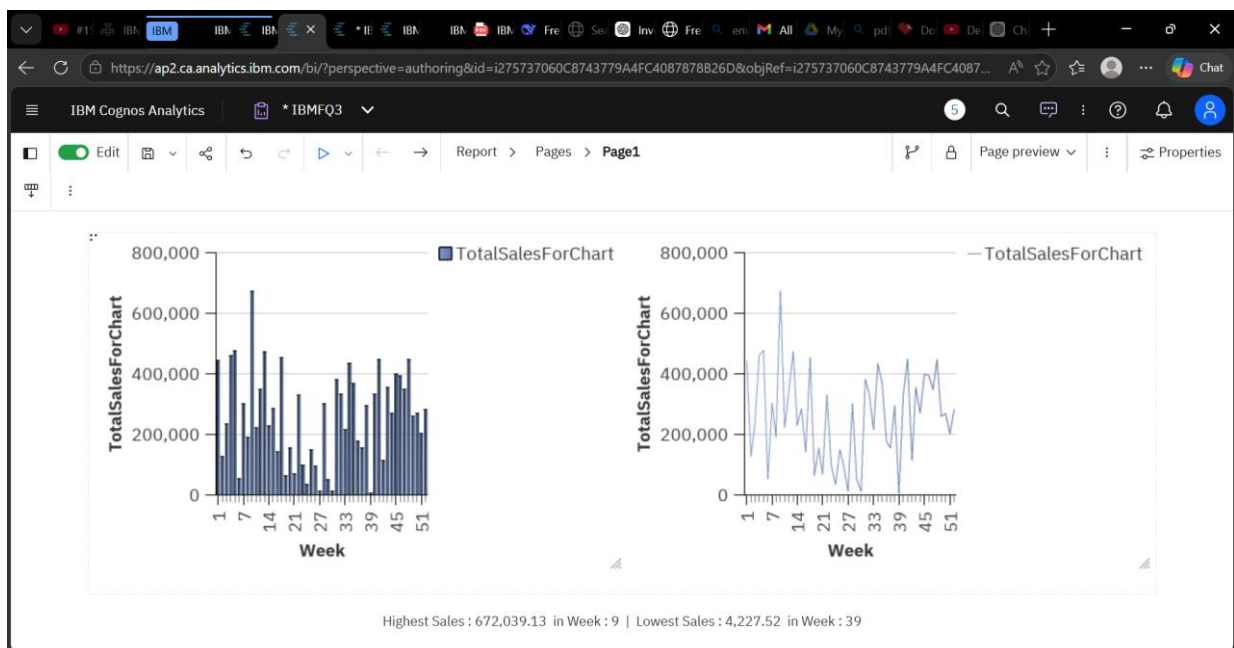
## Question 3 [Reporting]

Using Report Visualize Prompt, build a visualization that allows managers to choose between a Bar chart and a Line chart to view Weekly Sales Trends across all outlets. Summarize the highest and lowest sales weeks using a text object at the report bottom. Add Conditional Styling to weeks with sales below the monthly average.

### Screenshots

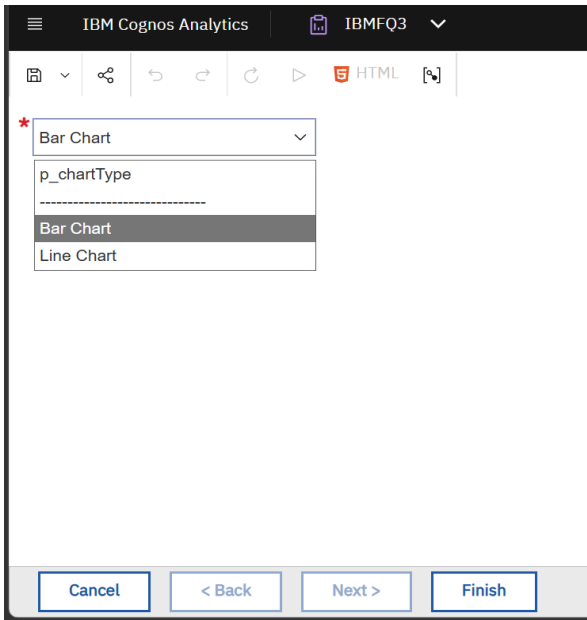
### *Weekly Sales Trend Analysis Report*

### Comparison of Weekly Sales Across All Outlets with Dynamic Chart Selection

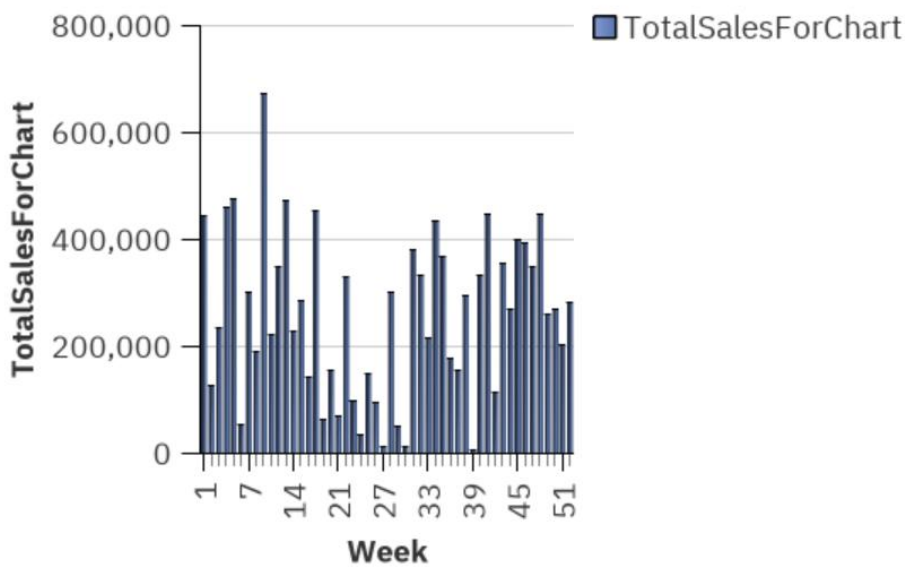


# Report Visualization Selection for Weekly Sales Analysis

Select Chart Type : Bar Chart



## Weekly Sales Trend (Bar Visualization)



Highest Sales : 672,039.13 in Week : 9 | Lowest Sales : 4,227.52 in Week : 39



# Select Chart Type : Line Chart

IBM Cognos Analytics

IBMFQ3

HTML

\*

p\_chartType

p\_chartType

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Bar Chart

Line Chart

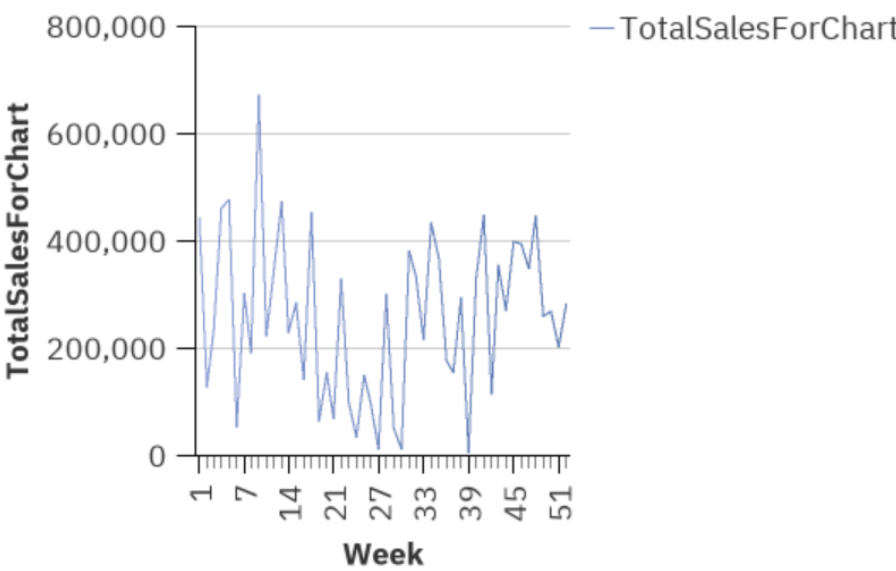
Cancel

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Next >

Finish

## Weekly Sales Trend (Line Visualization)

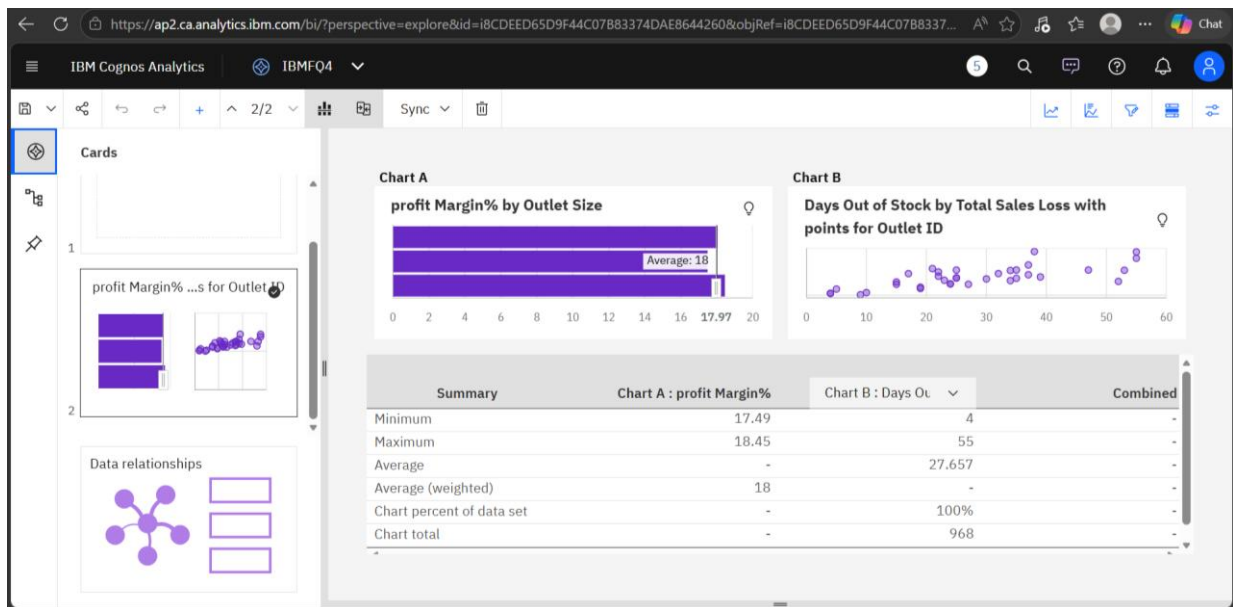


Highest Sales : 672,039.13 in Week : 9 | Lowest Sales : 4,227.52 in Week : 39

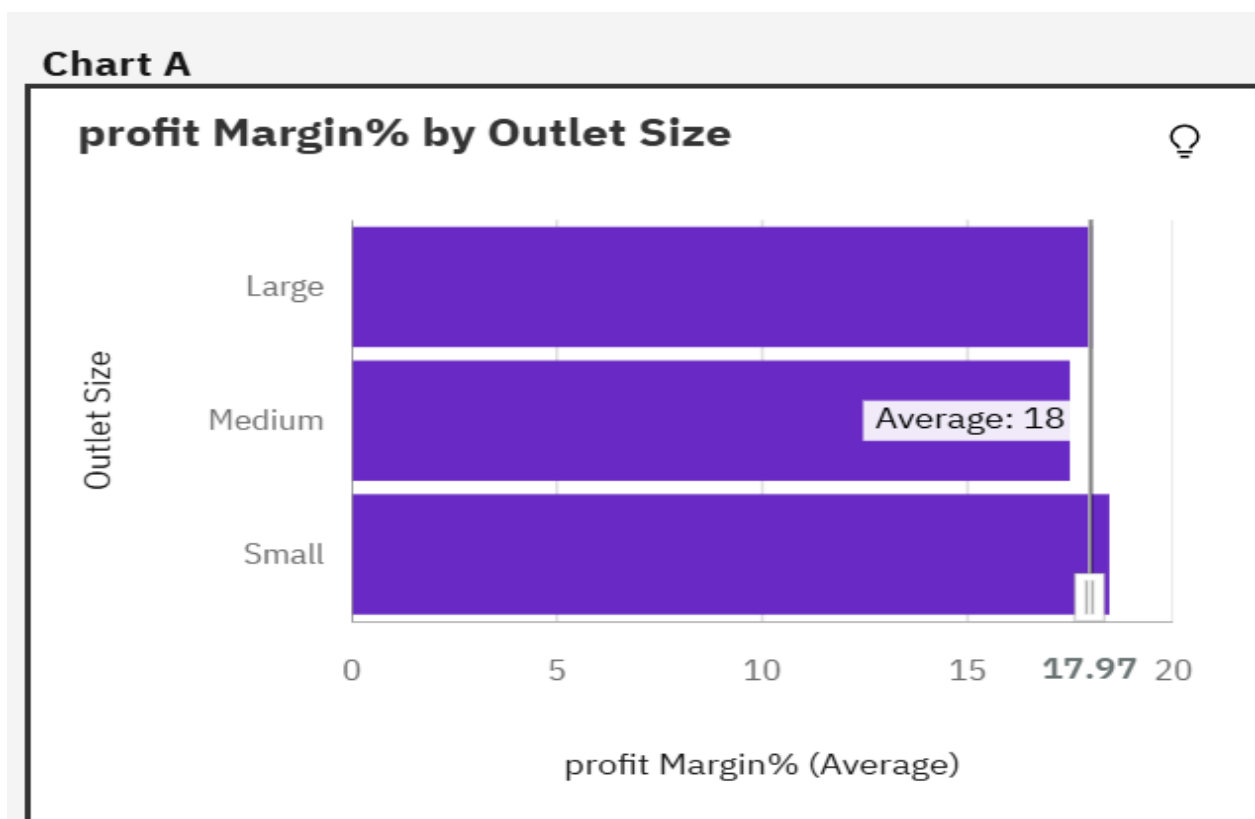
## Question 4 [Exploration]

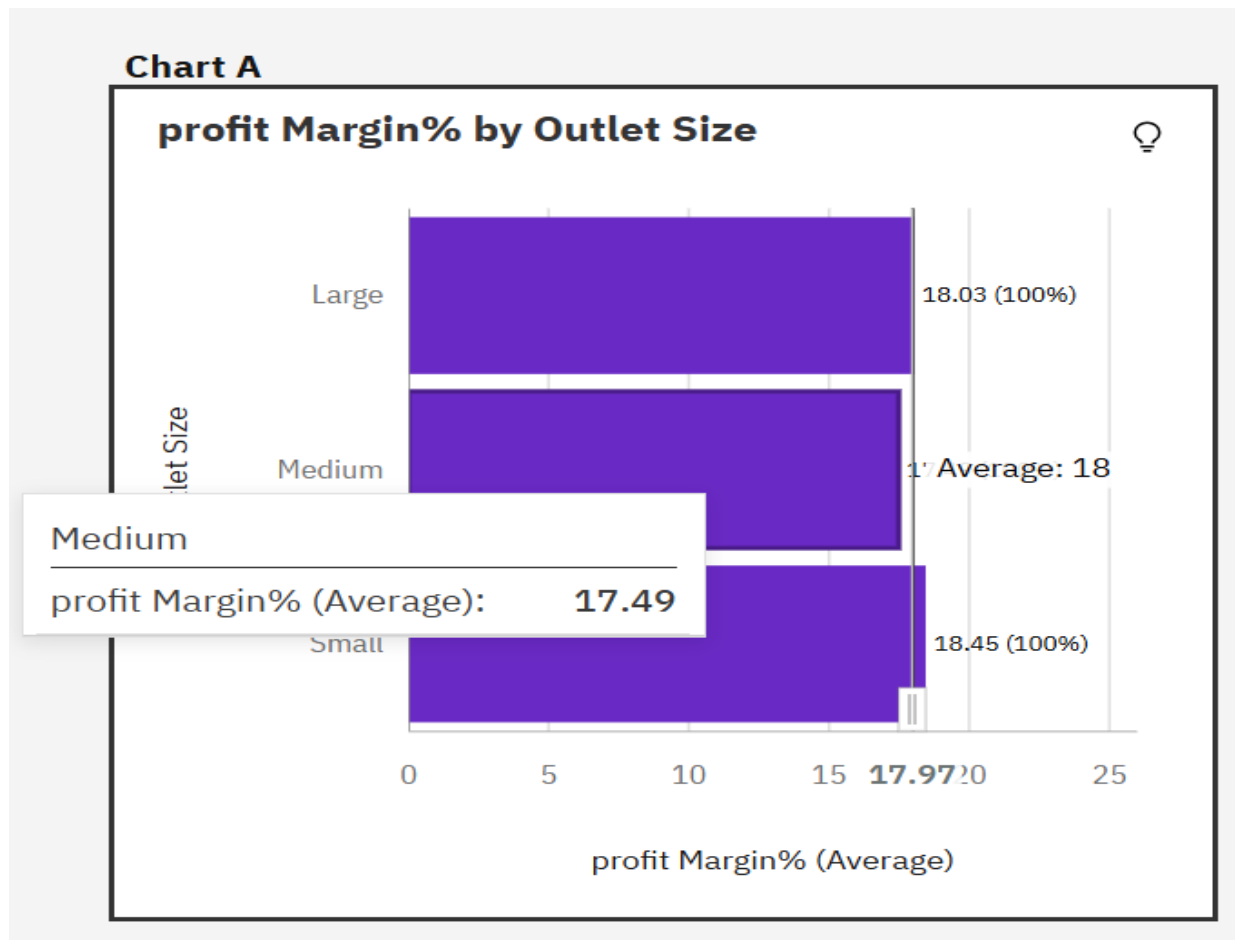
Using Cognos Exploration on the FreshMart dataset, explore the relationship between Outlet Size and Profit Margin. Investigate whether larger outlets consistently outperform smaller ones. Also explore how Days Out of Stock correlates with Total Sales loss. Capture and annotate all key exploration charts in your submission.

### Screenshots



### *Profit Margin % by Outlet Size*





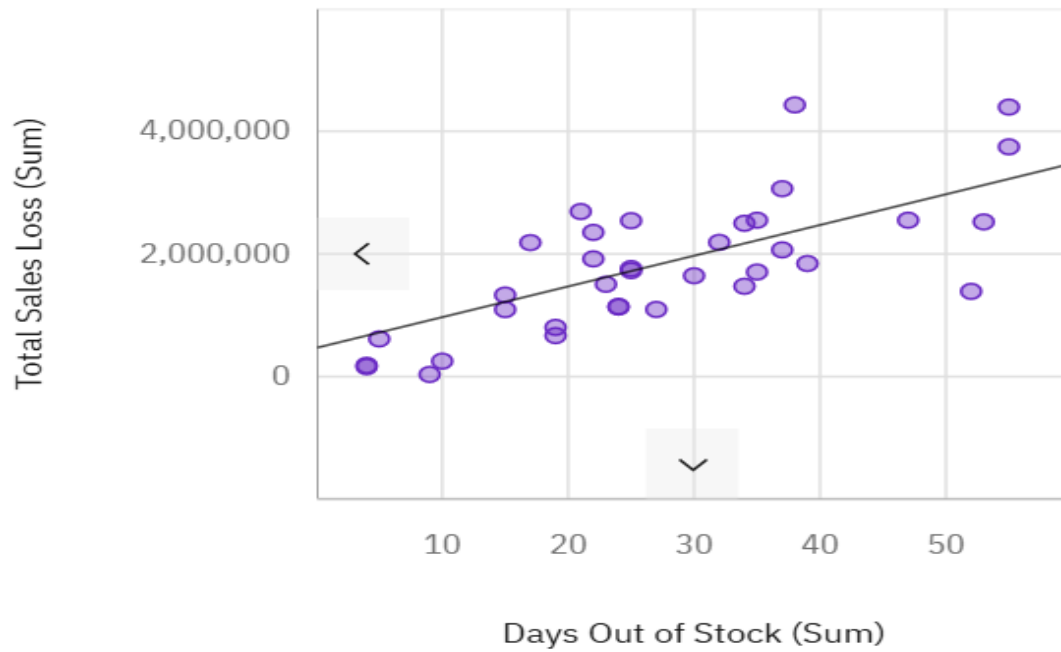
Outlet size has minimal impact on profit margin, with small outlets performing slightly better than others.

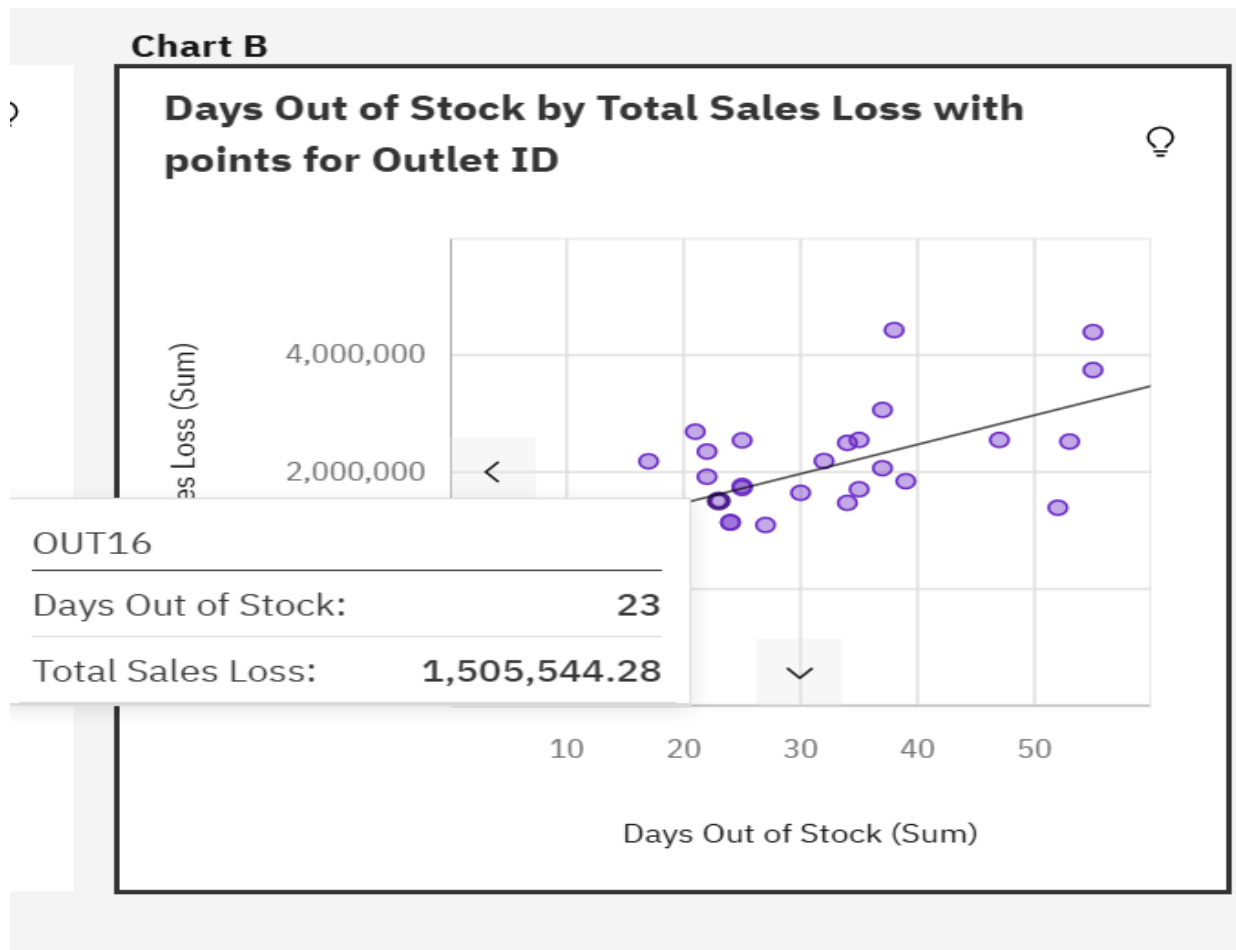
- Average Profit Margin = **18%**
- Small = **18.45% (Highest)**
- Medium = **17.49% (Lowest)**
- Small & Medium = **69.5% of total outlets**

## *Days Out of Stock vs Total Sales Loss (Scatter Plot)*

**Chart B**

**Days Out of Stock by Total Sales Loss with points for Outlet ID**





Poor inventory management directly leads to increased sales loss, making stock availability a critical factor.

- Total Sales Loss > **63 million**
- Positive relationship between variables
- Equation: Total Sales Loss = 468674 + 50063 × Days Out of Stock

## Question 6 [Story Making]

Create a Cognos Analytics Story for the FreshMart Quarterly Business Review with scenes:

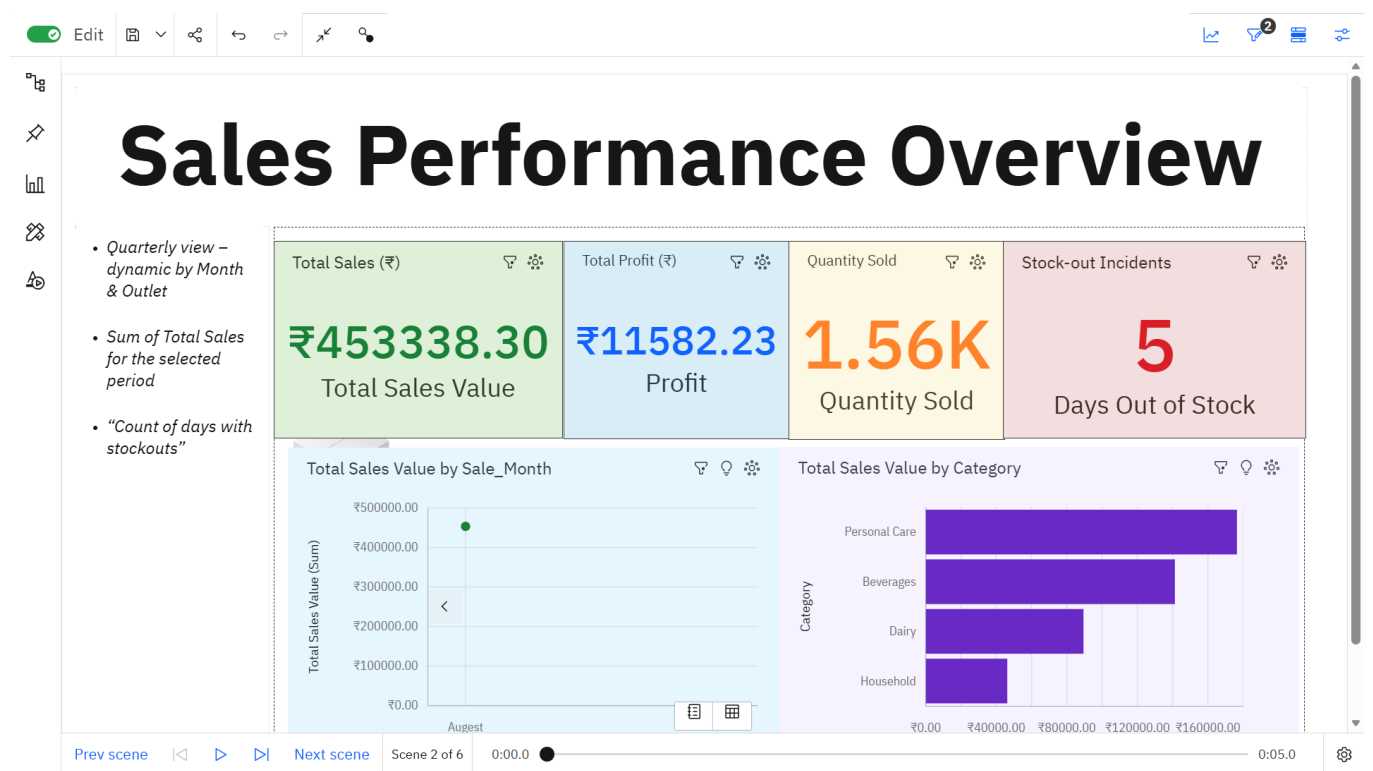
Scene 1 — Sales Performance Overview, Scene 2 — Top and Bottom Performing Outlets,

Scene 3 — Category Revenue Breakdown, Scene 4 — Inventory & Stockout Impact Analysis,

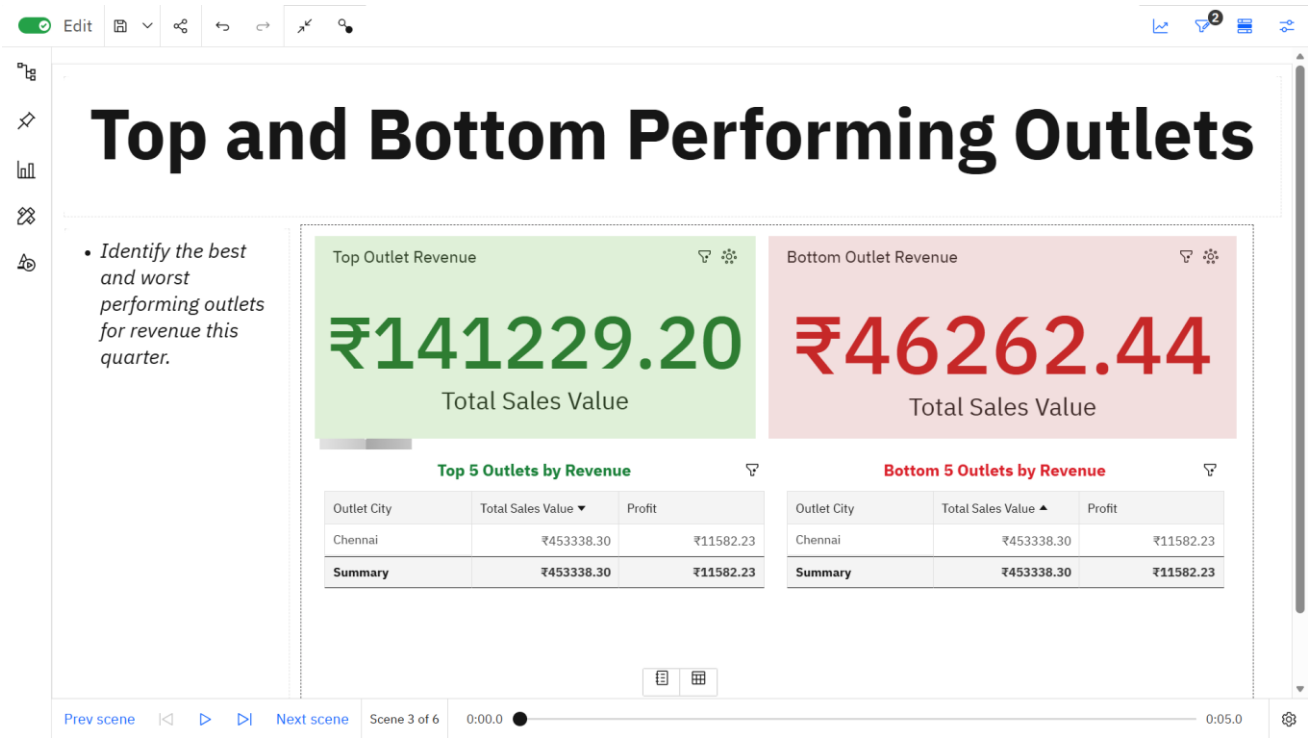
Scene 5 — Recommendations to Reduce Stockouts and Boost Slow Movers.

### Screenshots

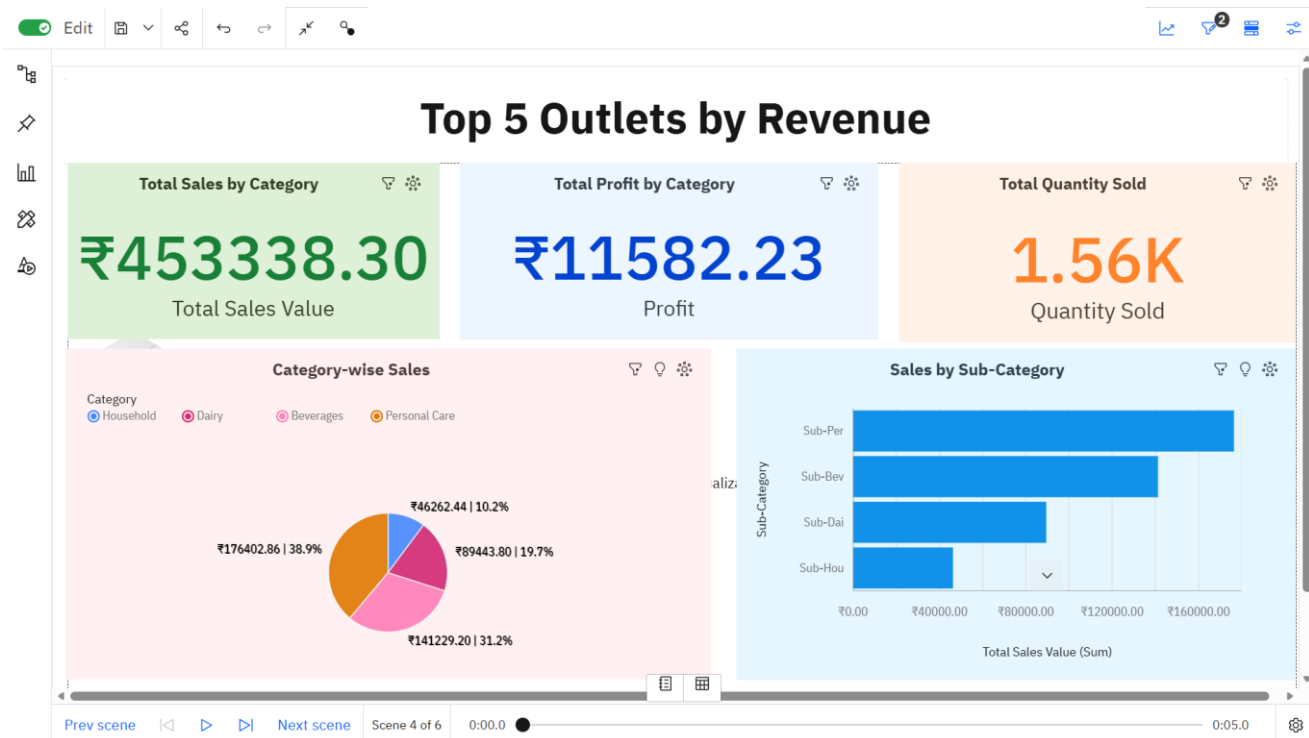
#### Scene 1 — Sales Performance Overview



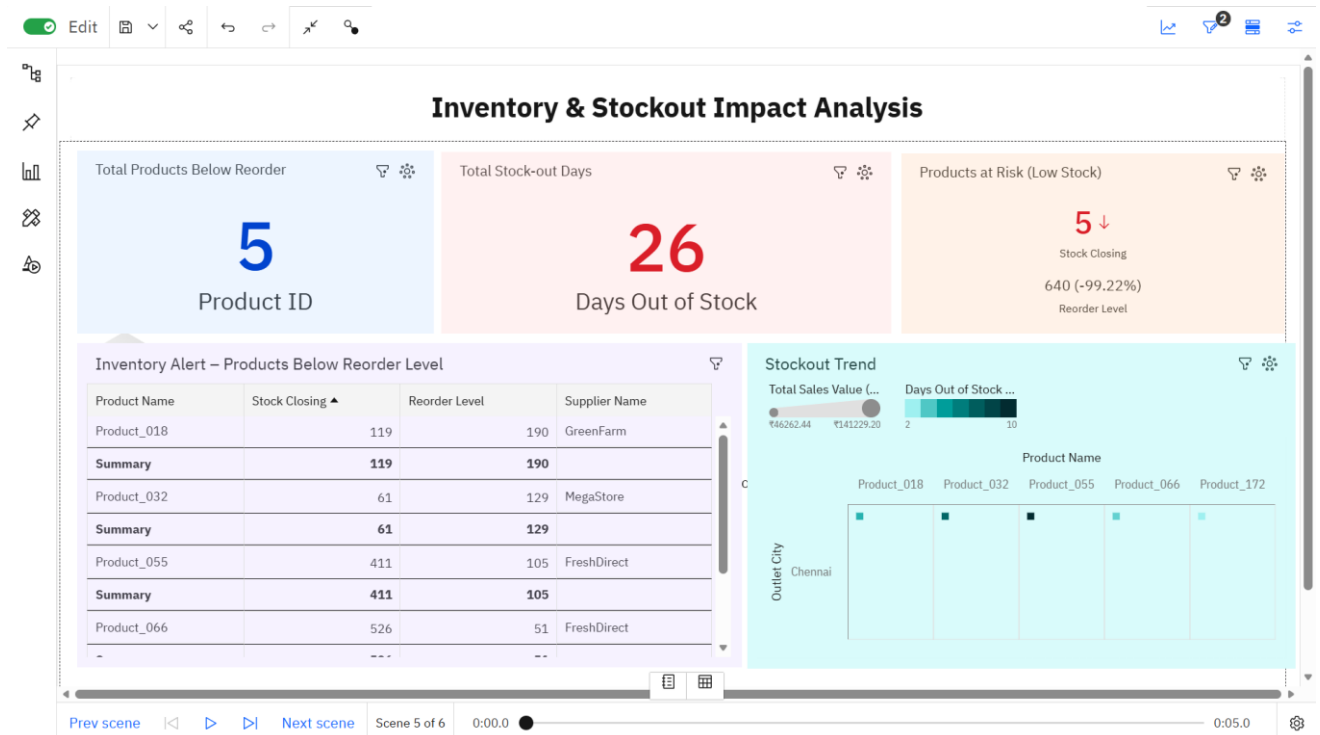
Scene 2 — Top and Bottom Performing Outlets



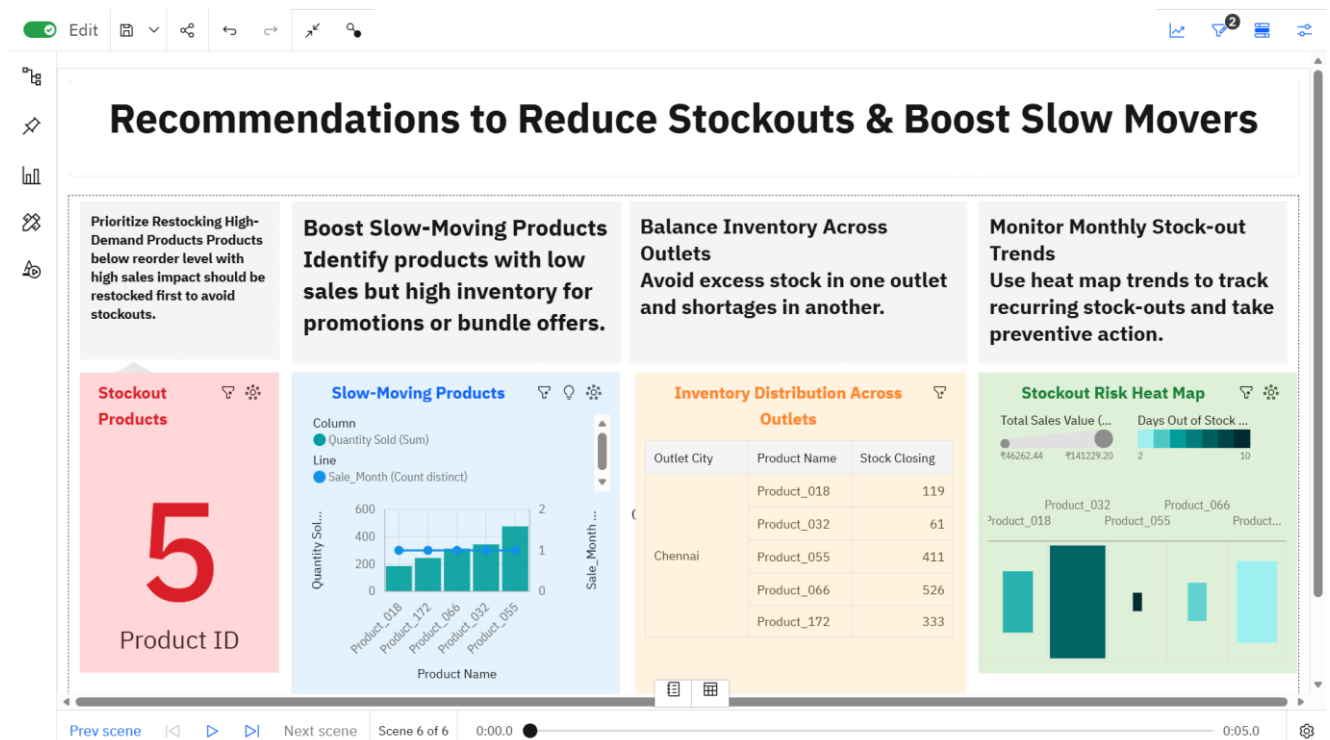
Scene 3 — Category Revenue Breakdown



## Scene 4 — Inventory & Stockout Impact Analysis



## Scene 5 — Recommendations to Reduce Stockouts and Boost Slow Movers



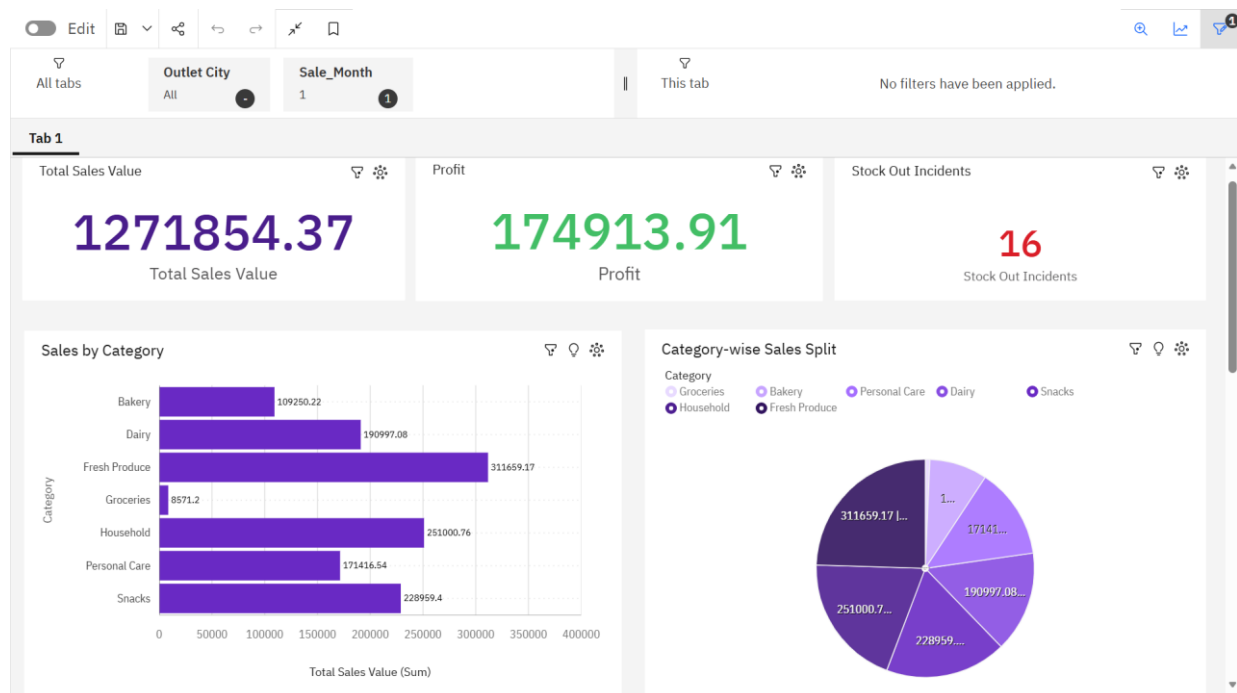


## Question 5 [Dashboard]

Design a Store Manager Dashboard containing: (1) KPI tiles — Total Sales, Total Profit, and Stock-out Incidents this month, (2) A Bar chart for Sales by Category, (3) A Map or table of Top 5 Outlets by Revenue, (4) A Pie chart for Category-wise Sales Split, (5) An Inventory Alert table showing products below reorder level. Add outlet and date filters.

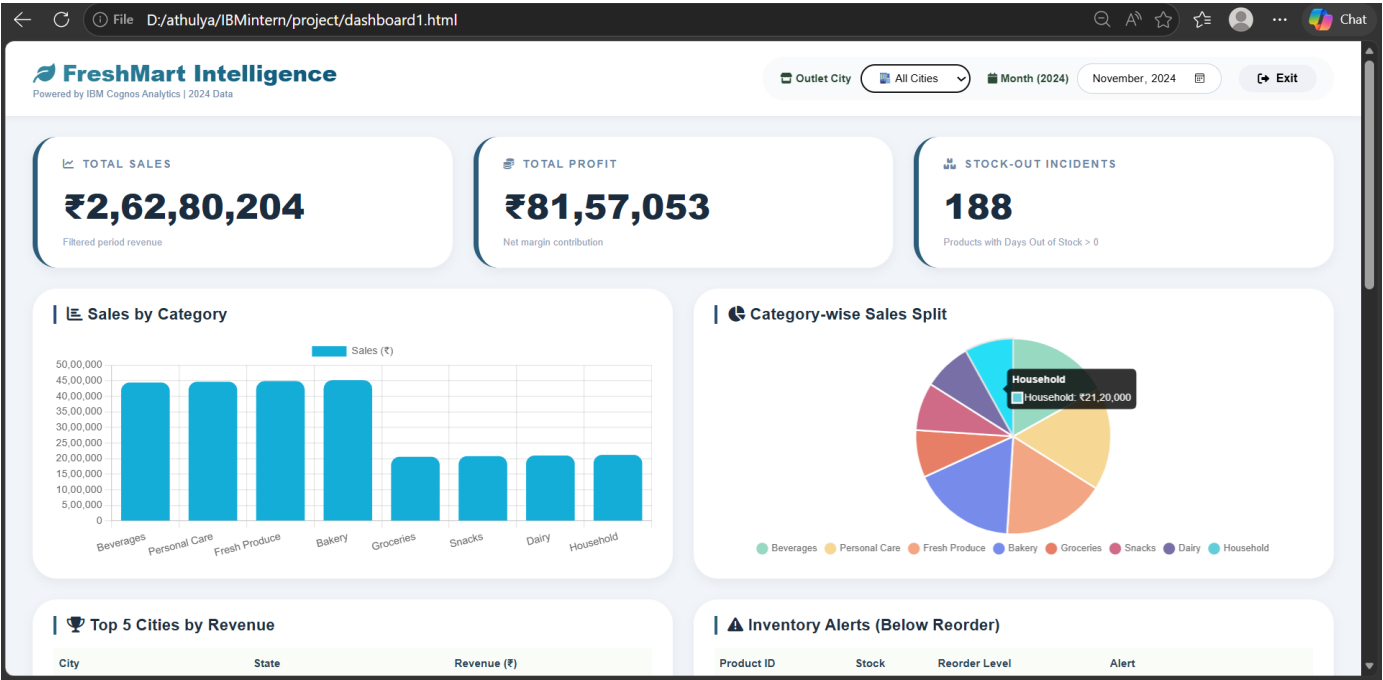
### Screenshots

## FreshMart Store Manager Dashboard - KPI Tiles, Category Charts, Outlet Rankings & Inventory Alerts





Dashboard Page:



# IBM Cognos | FreshMart Live Dashboard

